
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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ABSTRACT

A broad body of prior work has explored the computational complexity of deriving two of the most fundamental families of explanations for the predictions of machine learning models: (i) *sufficient reasons*, which are subsets of input features that, when fixed, determine a prediction, and (ii) *contrastive reasons*, which are subsets of input features that, when modified, alter a prediction. These explanations have been studied across a wide spectrum of settings — spanning, for example, non-probabilistic and probabilistic frameworks as well as local and global regimes — yet largely in isolation from one another. In this work, we introduce a single unifying framework that encompasses this entire landscape, showing that all of these explanation types can be characterized uniformly as the minimization of one probabilistic value function. We then prove that the complexity of computing them is governed by three general properties of this value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — three cornerstone properties of *combinatorial optimization*. Casting the problem at this level of generality uncovers several counterintuitive phenomena that hold broadly across the explanation settings we consider. For instance, although the *local* value functions exhibit none of these properties, we show that their *global* counterparts possess all of them. This distinction lets us establish a series of novel polynomial-time results for computing a wide variety of explanations with provable guarantees in the global explainability setting, applicable across essentially the entire interpretability spectrum of models — from transparent models such as decision trees to expressive black boxes such as neural networks and tree ensembles. Conversely, we show that even drastically simplified versions of these same explanations become NP-hard in the corresponding local setting, indicating that the gap we identify is a general feature of the local-versus-global divide rather than an artifact of any particular model or distribution.

1 INTRODUCTION

Despite substantial progress across the many methods proposed for explaining the decisions of machine learning models, the literature has repeatedly and unfortunately found that a wide range of desirable explanation types — spanning many different provable guarantees — are computationally hard to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), with this difficulty typically compounding as one moves toward more complex or highly non-linear model families (Barceló et al., 2020; Adolfi et al., 2025; 2024). As a result, the *computational complexity* of obtaining explanations has emerged as a central theoretical focus, with a large and growing body of work aiming to chart, across the broadest possible range of explanation types and model families, which explanations can be obtained efficiently and which remain fundamentally out of reach (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Among the many explanation notions studied through the lens of computational complexity, two families stand out as especially fundamental and broadly applicable across model types: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S such that when these features are fixed to specific values, the model’s prediction

remains unchanged, regardless of the values assigned to the complementary set $\bar{S}$. A *contrastive reason* is a subset of input features S such that modifying these features leads to a change in the model’s prediction.

Unlike additive attribution methods, which allocate importance scores across features but are often hard for humans to interpret (Kumar et al., 2020) or lack actionability (Bilodeau et al., 2024), sufficient and contrastive reasons provide *discrete, condition-like explanations* that directly answer “what is enough to justify this prediction?” or “what must change to flip it?”. This general and intuitive form of reasoning has given them a central place across a wide array of classic XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), has been shown to improve human prediction more effectively than additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and has proven valuable across a broad set of downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A well-established and widely shared principle further holds that *smaller* sufficient and contrastive explanations tend to enhance interpretability across settings, elevating *minimality* to a central and generally applicable guarantee of interest (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest complexity-theoretic studies of sufficient and contrastive reasons, and their findings already hinted at how sharply the difficulty can vary across model families. They established that finding the minimal-sized sufficient reason is NP-hard for decision trees and escalates to Σ_2^P -hard for neural networks. For contrastive reasons, they showed that computing the smallest explanation is solvable in polynomial time for decision trees yet becomes NP-hard for neural networks. Analogous hardness results were subsequently shown to extend to tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b), reinforcing that such barriers arise broadly across expressive model classes.

From non-probabilistic to probabilistic explanations. A common and recurring criticism of the classic definition of sufficient and contrastive reasons is its rigidity: because the guarantees must hold in an absolute sense over entire domains, they can readily yield excessively large or uninformative explanations across many models and tasks (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To address these limitations in a broadly applicable way, the literature has moved toward a more general definition that adopts a *probabilistic* view of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Within this framework, one seeks subsets of input features that influence a prediction with probability exceeding a given threshold δ , a relaxation that subsumes the absolute notions as a limiting case and applies uniformly across settings.

Wäldchen et al. (2021) were the first to study the complexity of probabilistic sufficient reasons, showing that for CNF classifiers the problem is already NP^{PP} -hard — and this hardness carries over broadly to richer model families such as tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A central and far-reaching theoretical insight in the probabilistic setting is the *lack of monotonicity* of the probability function (Arenas et al., 2022; Izza et al., 2023), a structural obstruction that makes even *subset minimal* explanations computationally hard in general. Strikingly, and underscoring how pervasive this obstruction is, (Arenas et al., 2022) show that finding a subset minimal probabilistic sufficient reason is NP-hard even for decision trees — in sharp contrast to the non-probabilistic case, where such explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. In a more recent study, Bassan et al. (2024) broaden the scope of complexity analysis from the *local* (non-probabilistic) setting — where explanations are tied to individual predictions — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons that characterize a model over entire domains. As with other non-probabilistic methods (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), however, the underlying criteria remain extremely strict — arguably even more so globally than locally, since the conditions must now hold everywhere. For instance, any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), which frequently forces the explanation to span nearly all input features and thereby limits its usefulness in practice.

OUR CONTRIBUTIONS

1. We unify a broad range of previously disparate explanation computation problems — sufficient and contrastive, probabilistic and non-probabilistic, local and global — within a single framework, cast as a minimization task over one general value function. Within this framework we isolate three general properties of the value function that fundamentally govern the complexity of the task across all of these settings: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.
2. We show that these properties behave in *strikingly different ways* depending on the structure of the value function, and that this difference holds uniformly across sufficient and contrastive reasons. In particular, we demonstrate the surprising and general result that while the *local* value functions for both explanation types are *non-monotonic*, their *global* counterparts are always *monotonic non-decreasing*. We further identify additional structural properties that hold throughout the global setting: the global sufficient value function is *supermodular*, whereas the global contrastive value function is *submodular* — neither of which holds in the local setting.
3. We leverage these properties to derive a broad set of new complexity results, revealing the general finding that global explanations with provable guarantees can be computed efficiently even where their local counterparts remain computationally hard. We instantiate these findings across three widely used model families that together span the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles — and, wherever possible, state them at the level of arbitrary models. First, we prove that while computing a subset-minimal local sufficient/contrastive probabilistic explanation is NP-hard even for decision trees (Arenas et al., 2022), its global counterpart is computable in polynomial time. We then extend this result to *any* black-box model (including complex classes such as neural networks and tree ensembles) under empirical distributions: obtaining a subset-minimal global sufficient/contrastive explanation is achievable in polynomial time, whereas the local version remains NP-hard throughout.
4. Finally, we establish an even stronger set of complexity results by exploiting the submodular and supermodular structure of the value functions in the global setting — structure that is entirely absent in the local case. We show that provable *constant-factor* approximation guarantees are attainable for computing cardinally minimal global explanations across arbitrary model classes — including complex ones such as neural networks and tree ensembles — whenever the empirical distribution is fixed. In sharp contrast, we prove strong inapproximability results for the local setting, showing that no *bounded* approximation is possible there even under drastically simplified assumptions.

Owing to space constraints, we present an overview of our main theorems and corollaries in the main text, stated at the broadest level of generality that our arguments support, with full proofs deferred to the appendix.

2 PRELIMINARIES

Setting. We work in a deliberately general setting. We consider an input space of arbitrary dimension $n \in \mathbb{N}$, with input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ may range over its own domain $\mathcal{X}_i$, which we allow to be either discrete or continuous, so that the full input feature space $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$ captures a wide variety of data types. We consider classification models $f : \mathbb{F} \rightarrow [c]$ for an arbitrary number of classes $c \in \mathbb{N}$. We also consider a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space, imposing no restrictions unless stated otherwise. Since accurately approximating $\mathcal{D}$ is computationally infeasible in many settings, a natural and widely applicable alternative is to work with a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$, which serves as a practical proxy for the “true” distribution — for instance, by taking $\mathbf{D}$ to be a sampled subset of the available training data.

The explanations we study come in two complementary flavors, local and global, which together cover the spectrum from instance-level to model-level reasoning. In the local case, an explanation targets a specific prediction $\mathbf{x} \in \mathbb{F}$, offering a form of reasoning for why the model predicted $f(\mathbf{x})$ on that instance. In the global case, an explanation instead aims to reflect the behavior of f across the

input space as a whole, independent of any particular $\mathbf{x}$, thereby characterizing the model’s overall decision-making logic.

Models. Many of our results are model-agnostic (e.g., the inherent properties of value functions) and thus apply to arbitrary classifiers; alongside these, we derive model-specific complexity results for several of the most widely used ML model families. To cover the interpretability spectrum as broadly as possible, we deliberately analyze models ranging from those typically regarded as “black-box” to those commonly considered “interpretable”. Specifically, we focus on: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, spanning both majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). Formal definitions are provided in the Appendix.

Distributions. In probabilistic explanation settings, complexity can depend heavily on the input distribution $\mathcal{D}$, so we study a spectrum of assumptions rather than committing to a single one. In particular, we consider three broad families of distributions: (i) *general distributions*, which impose no assumptions whatsoever on $\mathcal{D}$ and thus encompass every possible distribution; we rely on these primarily when proving properties that hold universally; (ii) *empirical distributions*, which include any distribution derived from a finite dataset $\mathbf{D}$ — an approach adopted very widely across XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which assume the features of $\mathcal{D}$ are mutually independent — another assumption in broad use throughout the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Importantly, empirical distributions do not entail feature independence; on the contrary, they can capture rich and complex dependencies extracted from finite data (Van den Broeck et al., 2022), which makes them a flexible modeling choice across many application scenarios. The complete formal definitions of these distributions are provided in the Appendix.

3 FORMS OF REASONS

In this section, we introduce the explanation types studied in this work. We begin with the strict non-probabilistic definitions of (local/global) sufficient and contrastive reasons and then broaden them into their more flexible and general probabilistic counterparts, which subsume the strict notions as special cases.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In the broad context of *feature selection*, users often wish to identify the top k features driving a model’s decision. We study the well-established *sufficiency* criterion for this selection, which underlies a wide range of commonly used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). Such feature selection can be performed either locally — focusing on a single prediction — or globally — across the entire input domain. Following standard conventions, we define a *local sufficient reason* as a subset of features $S \subseteq \{1, \dots, n\}$ such that when the features in S are fixed to their corresponding values in $\mathbf{x}$, the model’s prediction remains $f(\mathbf{x})$ regardless of the values assigned to the remaining features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff the following condition holds:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes a vector whose features in S take their values from $\mathbf{x}$ and whose features in $\bar{S}$ take their values from $\mathbf{z}$. This notion is closely connected to a broader family of explanation techniques: local sufficient reasons generalize *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which search for alternative inputs $\mathbf{z}'$ that keep the prediction unchanged (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while staying as “close” as possible to the original point. Instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and measuring proximity via Hamming distance makes the two notions coincide.

A *global sufficient reason* (Bassan et al., 2024) is a subset of input features $S \subseteq \{1, \dots, n\}$ that serves as a local sufficient reason for *every* possible input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A second, equally prevalent approach to explanation identifies subsets of input features that *modify* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024), aiming to pinpoint the minimal changes needed to alter the outcome. This perspective is complementary to sufficiency and applies just as broadly across models and tasks. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* concerning $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons likewise capture, as a special case, the widely studied family of *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ whose prediction flips, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. Taking the distance metric to be the Hamming weight makes the two notions coincide.

Just as with sufficient reasons, one can lift this notion to the global level, seeking a subset of features that, when altered, changes the prediction across all inputs in the domain of interest. This global form connects naturally to a broad range of practical settings, including approaches for identifying bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) as well as *group* counterfactual explanation methods that search for counterfactuals over many data instances at once (Carrizosa et al., 2024; Warren et al., 2024). Formally, we define a subset S as a *global contrastive reason* with respect to f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As discussed in the introduction, non-probabilistic sufficient and contrastive reasons are frequently criticized for imposing overly strict conditions, motivating a shift to *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021). These definitions generalize the original requirements by asking only that the guarantees hold with probability at least $\delta \in [0, 1]$, yielding a single flexible family that recovers the original non-probabilistic notions exactly in the limiting case $\delta = 1$.

Probabilistic sufficient reasons. We define $S \subseteq \{1, \dots, n\}$ to be a *local δ -sufficient reason* with respect to $\langle f, \mathbf{x} \rangle$ if, when the features in S are fixed to their values in $\mathbf{x}$ and the remaining features are sampled from an arbitrary distribution $\mathcal{D}$, the classification is preserved with probability at least δ . In other words:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ denotes that the features in S of vector $\mathbf{z}$ are fixed to their corresponding values in $\mathbf{x}$. Lifting this to the domain level, we adopt the standard notion of global explanations by averaging over all inputs: we define a subset $S \subseteq \{1, \dots, n\}$ to be a *global δ -sufficient reason* with respect to $\langle f \rangle$ if the expectation of the local sufficiency probability over samples drawn from $\mathcal{D}$ is at least δ . In other words:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. Mirroring the non-probabilistic case, we define a *local δ -contrastive reason* as a subset of input features that alters a prediction with sufficiently high probability. Here, in contrast to sufficient reasons, we fix the features of the *complementary* set $\bar{S}$ to their values in $\mathbf{x}$, and, allowing the features in S to vary, we require the prediction to differ from $f(\mathbf{x})$ with probability exceeding δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global setting, and fully analogous to global sufficient reasons, we define a subset S to be a *global δ -contrastive reason* by taking the expectation of the local contrastive probability over samples drawn from $\mathcal{D}$ and requiring this expected value to exceed δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

As discussed in the introduction, across every explanation form considered so far — sufficient or contrastive, local or global, probabilistic or not — the literature broadly agrees that explanations of *smaller size* are more meaningful, which makes their *minimality* a particularly important and widely sought provable guarantee. We accordingly study two central and general notions of minimality, both of which apply uniformly to all our explanation types:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

We note that cardinal-minimality is strictly stronger than subset-minimality: every cardinally minimal S is subset-minimal, but not conversely (see Appendix B.3 for an example). We deliberately use the terms subset- and cardinally minimal, rather than local and global minima, to avoid conflation with local vs. global explanations (input- vs. domain-level reasoning). Crucially, both notions apply to *every* explanation form we analyze, giving them the same broad reach as the framework itself. For instance, a cardinally minimal local probabilistic δ sufficient reason is the one with the smallest $|S|$, while a subset-minimal one is any S for which no proper subset $S' \subsetneq S$ also qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

A key observation is that the entire landscape of computational problems discussed so far — local or global, sufficient or contrastive, probabilistic or not — collapses into a single template: finding a minimal-size subset S such that $v(S) \geq \delta$, where the non-probabilistic settings correspond simply to $\delta := 1$. To make this precise, we introduce notation for the value functions: let v_{suff}^ℓ denote the local sufficiency probability from equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{suff}^g denote its global variant (equation 6). Analogously, let v_{con}^ℓ and v_{con}^g denote the local and global contrastive probabilities (equations 7 and 73, respectively). With this notation in hand, we can state a single unified task that captures all of these problems at once — that of finding a cardinally minimal δ -local/global sufficient/contrastive reason:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

In the same spirit, the relaxed goal of obtaining a subset-minimal — rather than cardinally-minimal — local or global sufficient/contrastive reason is captured by the following relaxed optimization objective, which again applies uniformly across all of our explanation forms:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subsetneq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. We now single out several general properties of the value function that, as we later show, play a decisive role in determining the complexity of generating explanations across the entire framework. The first is *non-decreasing monotonicity*, which guarantees that the marginal contribution $v(S \cup \{i\}) - v(S)$ is always non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining two properties, *supermodularity* and *submodularity*, likewise concern the behavior of the marginal contribution $v(S \cup \{i\}) - v(S)$, and together they cover the two canonical regimes

of diminishing versus increasing returns. In the supermodular case this contribution is itself a monotone non-decreasing function of the set, whereas in the dual *submodular* case it is a monotone non-increasing function. Formally:

Definition 3. *Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:*

1. *v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.*
2. *v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.*

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

Prior work shows that in the local setting of non-probabilistic explanations, subset-minimal sufficient or contrastive reasons can be computed thanks to monotonicity — but only in the narrow special case $\delta = 1$. This restriction is severely limiting and offers little practical flexibility. One might naturally hope to extend the tractability to probabilistic guarantees for arbitrary δ , yet a broad line of prior results shows that monotonicity breaks down throughout this more general setting, rendering the computation of explanations substantially harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

In the still more extreme *global* and *non-probabilistic* setting, Bassan et al. (2024) established a stringent *uniqueness* property — namely, that there is exactly *one* subset-minimal explanation. Requiring $\delta = 1$, however, makes this setting highly restrictive, all the more so globally, where the conditions must hold for *every* input. Indeed, Bassan et al. (2024) show that this unique minimal subset coincides with the set of all features that are not strictly redundant (which in practice can be nearly all of them). We prove that once one moves to the general global *probabilistic* setting — for arbitrary δ — this uniqueness property *fails* altogether, and the number of such subsets can in fact grow *exponentially*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Interestingly, although uniqueness collapses beyond the special case $\delta = 1$, the crucial *monotonicity* property survives in full generality: we show that it holds for the global value function at *every* value of δ , and for both the sufficient (v_{suff}^g) and contrastive (v_{con}^g) value functions alike. This is a surprising and broadly applicable finding, standing in sharp contrast to the local setting, where the corresponding value functions (v_{suff}^ℓ and v_{con}^ℓ) never satisfy it:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond the surprising insight that monotonicity holds globally yet fails locally, we uncover further structural properties that are unique to — and pervasive throughout — the global setting, namely submodularity and supermodularity. In particular, under the widely used assumption of feature independence, the global sufficient value function v_{suff}^g is supermodular. Its local counterpart v_{suff}^ℓ , by contrast, fails to be supermodular even under the far more restrictive assumption of a uniform (and hence independent) input distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

Interestingly, for the complementary family of explanation settings — that of obtaining a global probabilistic *contrastive* reason — we show that the corresponding value function is not supermodular but rather *submodular*, so that the two global settings together exhibit the full pair of dual structural properties. This is again especially striking against the local setting, where the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

In this section, we study the complexity of obtaining *subset minimal* explanations (local/global, sufficient/contrastive) across the full range of model types we analyze. The property that drives everything here is the *monotonicity* of the value function. As established in the previous section, monotonicity holds for both global value functions, v_{con}^g and v_{suff}^g , yet fails for both local value functions, v_{con}^ℓ and v_{suff}^ℓ . This single distinction is what determines whether a greedy algorithm converges to a subset minimal explanation: it does so throughout the global setting but not the local one. Consequently, we obtain the broadly applicable and surprising finding that computing subset-minimal explanations is hard across the local settings, whereas it is tractable (polynomial-time solvable) across their global counterparts. We begin with the following general greedy algorithm, which serves uniformly for all of these settings:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

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1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$   $\triangleright S$  is a (subset minimal?)  $\delta$ -explanation
```

Algorithm 1 is designed to produce a subset-minimal δ -explanation for any of the settings under study. It initializes the subset S to the full input space $\{1, \dots, n\}$. At each step it checks whether the smallest attainable value of $v(S \setminus \{i\})$ still exceeds δ , and if so removes the feature j that decreases the value function the least — that is, the one maximizing $v(S \setminus \{j\})$. When this process terminates, it returns S .

Whether Algorithm 1 yields a subset-minimal explanation hinges entirely on the *monotonicity* of the value function v . The algorithm terminates in a state where removing any single feature from S drives $v(S \setminus \{i\})$ below δ ; monotonicity then extends this to *any* $v(S \setminus S')$, a far stronger guarantee. Building on the monotonicity results established in the previous sections, we obtain the following proposition, which holds across all of our explanation settings:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this result, we now establish a broad set of new complexity findings for deriving subset-minimal explanations within our framework, spanning the various model types we analyze.

Decision trees. We begin with what is arguably the simplest and most transparent scenario one could hope for — taking f to be a decision tree and $\mathcal{D}$ to be an independent distribution — precisely because a separation here is the most striking. Even in this “interpretable” regime, Arenas et al. (2022) established the surprising intractability result that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions (indeed, even under the uniform distribution). In sharp contrast, we show the unexpected positive result that the very same problem becomes efficiently solvable in the global setting: a subset-minimal global δ -sufficient reason for decision trees can be computed in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless*

PTIME=NP, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.

Extension to other tractable models. The same tractability guarantee for computing global explanations extends beyond decision trees to the broader class of *orthogonal DNFs* (Crama & Hammer, 2011) — recall, Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. This extension is valuable precisely because orthogonal DNFs strictly *generalize* decision trees while retaining their clean structural properties, showing that our guarantees are not tied to tree-structured models but apply across a wider family of representations. For completeness, we give the full argument in Appendix N.

Neural networks, tree ensembles, and other complex models. We now push our results well beyond decision trees to arbitrarily complex model classes. Specifically, we show that whenever the distribution $\mathcal{D}$ is empirical and the model f admits polynomial-time inference — a mild and near-universal assumption — computing a subset-minimal global δ -sufficient or contrastive reason can be done in polynomial time, regardless of the internal structure of f .

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This strong complexity outcome — which holds for *any* model under an empirical distribution — lets us draw an even sharper line between the local and global regimes. Indeed, for expressive model classes, computing subset-minimal *local* explanations stays intractable even when restricted to empirical distributions. We establish this for both neural networks and tree ensembles, yielding the following theorem, which exhibits a strict complexity separation across these widely used model families:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn from subset-minimal reasons to the even more demanding task of computing a *cardinally minimal* δ -sufficient/contrastive reason. We show that throughout the global setting, the interplay of monotonicity with supermodularity/submodularity yields novel *provable approximation* guarantees for computing explanations, whereas in the local setting the same task remains intractable. This broadens the surprising global-versus-local divide from exact tractability to approximability, showing that it persists at multiple levels of the problem.

A unified greedy approximation algorithm. For a monotone non-decreasing submodular function, the task of finding a cardinally minimal explanation aligns closely with the classic *submodular set cover* problem (Wolsey, 1982), while for a *supermodular* function it corresponds to a non-submodular variant of the same problem (Shi et al., 2021). Both problems are well studied and come with strong, general approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). The standard tool for submodular set cover is a classic greedy algorithm, which we present next (Algorithm 2); it forms a single foundation for approximating a cardinally minimal sufficient or contrastive δ -reason, and we analyze its guarantees for each case in turn.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 closely mirrors Algorithm 1 but proceeds bottom-up rather than top-down: it starts from the empty subset and incrementally adds features, each time choosing the one that minimizes the

increase in $v(S \cup i)$, and halts once any further addition would push the value past δ . This shared structure lets a single procedure cover both explanation families.

Cardinally minimal contrastive reasons. We start with global contrastive reasons, where monotonicity and submodularity both hold, so that the task reduces exactly to the classic *submodular set cover* problem and Wolsey (1982)’s classic guarantee applies directly via Algorithm 2. For integer-valued objectives the algorithm attains a Harmonic-based factor, and more generally an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ with fixed sample size this becomes a *constant approximation* — only *logarithmic* in the sample size — a strong guarantee that holds across arbitrary model classes. In the *local* setting, by contrast, no *bounded* approximation exists at all, even for a *single* sample point, marking a sharp and general gap between the global and local cases.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We complement this with a matching *lower bound* for Theorem 3 that holds even in the general case of an empirical distribution without any independence assumption (see Appendix M), showing that the guarantee is essentially tight.

Cardinally minimal sufficient reasons. Whereas the submodular set cover problem — corresponding to cardinally minimal global *contrastive* reasons — admits strong approximations, the supermodular variant tied to global *sufficient* reasons is inherently harder to approximate. Nevertheless, it still admits general guarantees whenever the function has bounded curvature (Shi et al., 2021). Writing the *total curvature* of a function $v : 2^n \rightarrow \mathbb{R}$ as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$, and drawing on Shi et al. (2021), we show that Algorithm 2 attains an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

As before, under a fixed empirical distribution this approximation becomes constant. While contrastive reasons enjoy the tighter $\mathcal{O}(\ln(|D|))$ bound, sufficient reasons incur an additional $\frac{1}{1-k^f}$ factor — yet still admit a constant-factor approximation, so the positive result extends across both explanation families. In sharp contrast, the local variant remains inapproximable, with no *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Taken together, these findings reinforce and broaden Subsection 7.1, which established the tractability of subset-minimal global explanations against the hardness of their local counterparts. We have now shown that the same divide extends to approximation: cardinally minimal global explanations are tractably approximable across model classes, whereas their local counterparts remain fundamentally inapproximable.

8 LIMITATIONS AND FUTURE WORK

Many of our central findings — especially those concerning the fundamental properties of value functions — are model- and distribution-agnostic and thus apply in broad generality; alongside them, we instantiate the framework to obtain concrete complexity results for particular model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions. As with any framework of this scope, further settings naturally remain open for analysis. We view this breadth as a strength: our results offer foundational insight into the nature of explanations together with new tractability and intractability guarantees, laying a general foundation on which a wide range of additional explainability scenarios can be studied in future work.

More specifically, our study opens up several broad and important theoretical directions:

1. First, it would be valuable to explore how far the tractable results we obtain for global explainability can be sharpened across the range of simpler model classes — decision trees, linear models, and other models with inherently tractable structure. Even though local explanations for such models are frequently intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms studied here carry strong structural properties that already yield markedly improved guarantees and may well admit further tightening. Improving the approximation guarantees under empirical distributions is likewise a natural and broadly relevant direction.
2. Second, when *exact* computation of the expectations is infeasible, a general question is whether other techniques — such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes like DNF formulas (Meel et al., 2019) (and hence tree ensembles), or Monte Carlo-based approximations of the kind used in (Subercaseaux et al., 2025) — can approximate them effectively across settings. A central challenge is understanding how to preserve minimality guarantees under approximation error, and identifying the tolerance levels for which they continue to hold.
3. Finally, we conjecture that a range of alternative notions of feature importance — e.g., other loss measures such as KL divergence (Conmy et al., 2023) — may exhibit the same patterns of monotonicity, submodularity, or supermodularity when moving from local to global explanations. If so, the efficient feature-selection guarantees we develop here would extend naturally to these broader metrics as well.

9 CONCLUSION

We have introduced a single unified framework that spans a diverse landscape of explanations and, through it, exposed a stark and general contrast between local and global sufficient and contrastive reasons. While the *local* variants possess none of the structural forms that make combinatorial optimization tractable — neither monotonicity, submodularity, nor supermodularity — we prove that their *global* counterparts exhibit exactly these properties throughout: (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These structural results underpin a series of surprising complexity findings, establishing that global explanations with provable guarantees can be computed efficiently across model classes as expressive as neural networks and tree ensembles. In sharp contrast, computing the corresponding local explanations stays NP-hard even in drastically simplified scenarios. Altogether, our results uncover foundational and broadly applicable properties of explanations and chart both the tractable and intractable frontiers of the field, opening a wide range of avenues for future research.

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Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work sits within the broad line of research on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A recurring theme across this area is the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because obtaining formal explanations is frequently intractable for the most expressive model families — such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a) — a substantial portion of the literature has concentrated on more tractable classes (Marques-Silva & Ignatiev, 2023), spanning decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic study, considerable effort has gone into computing formal explanations in practice via automated reasoning tools across many model families — for example, MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for deriving certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). A further body of work seeks to alleviate the underlying computational burden through a variety of general strategies, including abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), and training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, a closely related strand within formal XAI studies methods that harness combinatorial and submodular optimization to improve the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), typically for decision tree classifiers. As that literature notes — and as we confirm here — the value-function objective underlying these local variants is not, in general, submodular. Greedy algorithms inspired by submodular optimization nonetheless serve as effective heuristics and often yield high-quality explanations empirically. Our work instead establishes a rigorous theoretical bridge between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons. This connection delivers formal guarantees and, more broadly, opens the door to a wide range of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

In this section, we formalize the full range of models and distributions referenced throughout the paper. Subsection B.1 defines the model families we consider, and Subsection B.2 formalizes the distributional assumptions.

B.1 MODEL FORMALIZATIONS

In this subsection, we formalize the three base-model families analyzed throughout the paper, chosen to span a broad portion of the interpretability spectrum: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph that represents a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes. The graph encodes the function as follows: (i) Each internal node v is assigned a distinct binary input feature from the set $\{1, \dots, n\}$; (ii) Every internal node v has at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) Along any path α in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from $[c]$. Thus, assigning values to the inputs $\mathbf{x} \in \mathbb{F}$ uniquely determines a path α from the root to a leaf in the DT, where the function output $f(\mathbf{x})$ corresponds to a class label $i \in [c]$. The size of the DT, denoted $|f|$, is defined as the total number of edges in the graph. To accommodate flexible modeling across a wide variety of trees, the ordering of input variables $\{1, \dots, n\}$ is allowed to differ across distinct paths α and α' , subject only to the constraint that no variable repeats along a single path.

Neural Networks. We state our *hardness* proofs for neural networks with ReLU activations, while our *tractability results* (i.e., polynomial-time algorithms) hold far more broadly, for any architecture admitting polynomial-time inference — a standard and near-universal assumption. As a result, every separation we establish between tractable and intractable cases carries over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, thereby covering a wide swath of models in common use, including ResNets, CNNs, Transformers, Diffusion models, and beyond. Note further that any ReLU network can be recast as a fully connected network by assigning zero weights and biases to the missing connections. Following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we therefore assume without loss of generality that the network is fully connected — that is, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f consists of $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and one output layer $g^{(t)}$, where each layer is defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$ denotes the weight matrix, $b^{(j)}$ the bias vector, and $\sigma^{(j)}$ the activation function of the j -th layer. The model thus comprises, in full generality, t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, representing the binary input vector. The network’s dimensions are governed by a sequence of positive integers $d_0, \dots, d_t$, with weight matrices and bias vectors given by $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$, respectively, all learned during training. Since the model acts as a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers use the ReLU activation $\text{ReLU}(x) = \max(0, x)$. Although a sigmoid activation is typically employed during training, for post-hoc interpretability we take the output layer to apply a threshold-based step function, $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise — a standard modeling choice that does not restrict the generality of our analysis.

Tree Ensembles. A great many ensemble methods exist; since our concern is post-hoc explanation, we focus on the *inference* phase rather than training. Our analysis is framed around the two dominant inference schemes — *majority voting* (e.g., Random Forests) and *weighted voting* (e.g., XGBoost). As with neural networks, however, our *tractability results* — the polynomial-time algorithms — are not confined to these two schemes: they extend to *any* ensemble configuration admitting polynomial-time inference, thereby covering a considerably broader range of ensemble designs.

Majority Voting Inference. In *majority voting* inference, for any number of classes c , the final prediction $f(\mathbf{x})$ is the class $j \in [c]$ receiving the majority of votes among the individual predictions $f_i(\mathbf{x})$ across all $i \in [k]$ (i.e., across each tree in the ensemble).

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ denotes the most frequent label in the multiset S . Ties are resolved by any fixed tie-breaking rule (e.g., lexicographic order or a predefined priority), and the choice of rule does not affect our results.

Weighted Voting Inference. In *weighted voting* inference, each ensemble member is assigned a weight $\phi_i \in \mathbb{Q}$ reflecting its relative importance, and the predicted class is the one accumulating the highest total weight across all members. Formally, for any $\mathbf{x} \in \mathbb{F}$, we define f as:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the full spectrum of distributional assumptions discussed in the main paper.

Empirical Distributions. The distribution $\mathcal{D}$ over the input features is defined directly from a dataset $\mathbf{D}$ comprising inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$, so that the probability of any given input $\mathbf{x} \in \mathbb{F}$ is simply its frequency of occurrence within $\mathbf{D}$:

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Formally, given a probability value $p(\mathbf{x}_i) \in [0, 1]$ for each individual input feature, we call the distribution $\mathcal{D}$ independent if the joint probability factorizes as $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

This family already subsumes a familiar special case: when $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, the distribution collapses to the *uniform distribution*.

General Distributions. Many of the proofs in this work — especially those establishing fundamental properties of value functions — hold under fully general distributional assumptions, and thus apply well beyond any single modeling choice. Empirical distributions (using the training dataset as a proxy) are the most common choice in XAI, but the framework accommodates a much wider array of options for approximating a distribution: k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for modeling tabular dependencies (Größer & Okhrin, 2022), or richer conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). Such choices recur throughout the counterfactual explanation and algorithmic recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which is closely related to contrastive explanations. Moreover, when the value function cannot be computed directly from a structural property of the model (e.g., leaf enumeration in decision trees) or from an empirical distribution, it can instead be approximated by general-purpose methods such as Monte Carlo sampling (Hastings, 1970), along the lines of (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

In this subsection, we discuss in more detail the distinction between cardinal and subset minimality, which applies uniformly to all explanation forms in our framework. Cardinal minimality is the

substantially stronger of the two guarantees, corresponding to a globally minimal explanation size, whereas subset minimality guarantees only a local form of minimality.

To see why cardinal minimality is stronger, observe that if $S \subseteq [n]$ is a cardinally minimal explanation, then no smaller set S' with $|S'| < |S|$ can be an explanation. Hence no strict subset $S' \subsetneq S$ is an explanation, so S is subset minimal as well.

The converse, however, fails in general. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment $\mathbf{x} := (1, 1, 1)$, for which $f(1, 1, 1) = 1$. Fixing feature $\mathbf{x}_1 = 1$ yields a cardinally minimal (and hence subset minimal) sufficient reason, since the prediction stays 1 regardless of $\mathbf{x}_2, \mathbf{x}_3$. Yet fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also yields a subset minimal explanation — one that is not cardinally minimal. This confirms that, in general, every cardinally minimal explanation is subset minimal but not conversely.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024), a phenomenon exhibited by functions of the following form. We show that the very same function also admits an exponential number of *global* and *probabilistic* sufficient reasons, so that the blow-up is not particular to the local non-probabilistic regime. This function is, moreover, a representative of the broad class of *threshold Boolean functions* of Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, it is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. The function f is *symmetric* in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends only on the number of 1's (equivalently, 0's) in the input and is invariant under any permutation of the input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, it can be verified that for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and also set $k := \lfloor \frac{n}{2} \rfloor$. Each such subset is *subset-minimal*, since any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). The same counting analysis as in (Bassan et al., 2024) therefore applies directly, giving exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Applying Stirling's approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the stated bound on the number of subset-minimal global δ -sufficient reasons, confirming that the exponential blow-up carries over to the global probabilistic setting.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^l and v_{suff}^l are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. In this section, we prove the monotonicity property for both types of global explanations — sufficient and contrastive — and in full distributional generality. We begin with global sufficiency. To build intuition, we first treat the simplest case of Boolean functions, then progressively broaden the result to functions with discrete multi-valued input domains and multiple output classes. We further extend it to continuous input domains, ultimately showing that monotonicity holds for *any* well-defined classification function. Finally, we transfer the same guarantees to global contrastive explanations. Crucially, none of these proofs invoke feature independence: monotonicity holds in a strong sense for *every* underlying distribution.

Lastly, we give a simple example showing that, by contrast, monotonicity fails for the local probabilistic sufficient and contrastive value functions.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\bar{S} \cup \{i\}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\bar{S} \cup \{i\}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\bar{S} \cup \{i\}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

To simplify notation, we occasionally omit the distribution notation $\sim \mathcal{D}$ in the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and use f^+ and f^- to denote the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We begin with a technical simplification of the probability term to facilitate the proof, and then proceed to establish monotonicity.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to the set S . Given a point $\mathbf{x} \sim \mathcal{D}$ and a set S , consider all points $\mathbf{z} \sim \mathcal{D}$ such that $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$ — they all share the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ that shares the same x_S and the same output b , we marginalize over $x_{\bar{S}}$. This yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, from which we can infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2],
\end{aligned} \tag{22}$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. We now proceed to prove the monotonicity claim, building on the previous simplification. We begin by introducing a few additional notations. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \tag{26}$$

Note that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and similarly: $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Using these identities along with the previous result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby concluding our proof. $\square$

Lemma 2. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each consist of a finite set of values. Let $\mathcal{K}$ represent the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\ \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \end{aligned} \tag{27}$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] &= \\ = \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \tag{28}$$

Let P_k denote $\mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Hence, $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which completes the proof for this setting. $\square$

Lemma 3. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend the monotonicity results to functions where features are defined over continuous domains. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in lemma 2 by revising the notation accordingly. The core reasoning remains unchanged when transitioning from the discrete to the continuous case—only the notation differs.

Alternatively, we provide a direct proof below. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Moreover, let $p(x_i)$ denote the marginal probability density function of x_i , and $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. Combining lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for any well-defined classification function — with Boolean, discrete, continuous, or mixed features — and under any data distribution.

$$v_{con}^g(S) = 1 - v_{suff}^g(\bar{S}) \quad (38)$$

This shows that v_{con}^g is monotonic non-increasing as features are added to $\bar{S}$ or, equivalently, monotonic non-decreasing as features are removed from $\bar{S}$ and added to S . The monotonicity guarantee therefore holds identically for both global explanation families, completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section establishes the supermodularity property for the global sufficient setting (from which the submodularity of the global contrastive setting follows). As with monotonicity, we

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

begin with Boolean functions and progressively broaden the argument to multi-valued discrete and continuous input domains, and finally to arbitrary well-defined classification functions. Unlike monotonicity, however, these proofs do rely on feature independence, and we complement them with a counterexample showing that supermodularity can indeed fail once that assumption is dropped — thereby delineating precisely how far the property extends.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$. That is, $\Delta_t(S, f)$ is monotone.

Clearly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ &\quad \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ &\quad \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ &\quad \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote

$\Pr(z_i = v'_k)$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v'_k, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right]. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right]. \end{aligned} \quad (45)$$

Therefore, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\ &\geq 0. \end{aligned} \quad (46)$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^{δ} is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. The supermodularity results of Lemma 5 extend to the continuous case, again assuming $\mathcal{D}_p$ satisfies feature independence. One route is to partition the domain of the continuous features into two disjoint subdomains and adapt the proofs of Lemma 5 by revising the notation accordingly.

For completeness, we instead give a direct proof below. Under feature independence, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t \mid \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let $Y_j(i, t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
\Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\
&= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}.
\end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned}
\Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \tag{50} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S,
\end{aligned}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\
&= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}.
\end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned}
\Delta_t(S', f) - \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{51}$$

Consequently, the global sufficient value function v_{suff}^g retains supermodularity for *any* valid classification function, irrespective of whether its features are Boolean, discrete, continuous, or a mixture thereof.

We now show that, in stark contrast, the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \tag{52}$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4},$$

$$v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. $\square$

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once the global sufficient setting is known to be supermodular, the submodularity of the global contrastive setting follows immediately by duality. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. Comparing the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$ is equivalent to comparing $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, concluding this part of the proof.

We now show that, once again in sharp contrast, the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the remainder of this section, we show that feature independence is not merely sufficient but in fact necessary for supermodularity, thereby pinning down the exact boundary of the property. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) + \\ & \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) \\ & = (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} + \\ & \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\ & = \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) \end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned} & \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1 \end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ & = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\ & \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\ & \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1) \end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned} & \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\ & = 1 \end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned} & (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\ & = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2, \end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned} & \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\ & = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2. \end{aligned}$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This gives $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, showing that — unlike monotonicity, which required no such assumption — feature independence is genuinely necessary for supermodularity to hold.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. This proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (Proposition 2), together with standard conclusions from prior work. Below we spell out how monotonicity yields the stated corollary uniformly across both explanation families.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g is a direct consequence of monotonicity: prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) shows that this style of greedy algorithm produces a subset-minimal explanation whenever the underlying value function is monotone non-decreasing. The algorithm halts once, for every $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , so that although $v(S) \geq \delta$ (a maintained invariant), removing any single element breaks the condition. Monotonicity then extends this to *every* proper subset of S , guaranteeing that S is subset-minimal.

We now explain why Algorithm 1 fails to converge to a subset-minimal solution in the local setting. It is well known that the local probabilistic sufficient value function (and likewise its contrastive counterpart) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution over the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

can be carried out in polynomial time via the following procedure. Exploiting the tractable structure of decision trees, we iterate over all pairs of leaf nodes, each corresponding to partial assignments $\mathbf{x}_S$ and $\mathbf{z}_S$. For each pair whose two leaves yield the same prediction under f , we compute the

corresponding term of the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the feature-wise probabilities over the shared features of the two vectors. Under feature independence these probabilities factor cleanly, so the full expectation is obtained by summing the contributions of all matching leaf pairs.

Since every probability involved is given as part of the input and each step performs only polynomial-time operations, the whole procedure runs in polynomial time. Hence each step of Algorithm 1 is efficient, and so is the algorithm overall. Combining this with Lemma 2, which shows that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation: the algorithm halts once, for all $i \in S$, the value $v(S \setminus \{i\})$ falls below δ while $v(S) \geq \delta$ is preserved, and monotonicity extends this to every strict subset of S , ensuring S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), who showed that, assuming $P \neq NP$, no polynomial-time algorithm computes a subset-minimal δ -sufficient reason for decision trees. Although this was proven specifically for the uniform distribution, the hardness immediately extends to the broader class of independent distributions, which contains it. Together, these results yield the full complexity separation between the local and global variants, concluding the proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

can be performed in polynomial time for *any* model whenever $\mathcal{D}$ is drawn from the class of empirical distributions. It suffices to iterate over pairs of instances $\mathbf{x}, \mathbf{z}$ in the dataset and run inference $f(\mathbf{x})$ and $f(\mathbf{z})$ to accumulate the expected value. Determining whether the probability function meets a given threshold δ is therefore also polynomial. Combining this with the non-decreasing monotonicity of the global probability function (Proposition D), we can apply Algorithm 1 for either contrastive or sufficient reason search. Since the algorithm issues only a linear number of these polynomial queries, a subset-minimal explanation — contrastive or sufficient alike — is obtained in polynomial time, and the argument applies to arbitrary models.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $P = NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part follows immediately from Proposition 6, which establishes the claim for arbitrary models, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ are polynomial-time computable under an empirical distribution. For the second part, we prove two lemmas, covering the intractability of the sufficient and contrastive cases respectively.

Lemma 6. *Unless $P = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We prove the claim first for neural networks and then extend it to tree ensembles. The argument shows that a polynomial-time algorithm for computing a subset-minimal local δ sufficient reason for neural networks would, in turn, solve the classic NP-complete *CNF-SAT* problem in polynomial time. *CNF-SAT* is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: *Yes*, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and *No* otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We in fact establish hardness for a much simpler, more restricted case, which then implies hardness in the general setting a fortiori. Here we let the empirical distribution $\mathcal{D}$ consist of a single element, which for simplicity we take to be the zero vector $\mathbf{0}_n$. Because the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single point $\mathbf{0}_n$, the input format (discrete, continuous, etc.) is immaterial, so the hardness result holds uniformly across all such settings.

As with $\mathbf{0}_n$, let $\mathbf{1}_n$ denote the n -dimensional all-ones vector. Given an input ϕ , we first evaluate ϕ at $\mathbf{1}_n$, i.e. with all variables set to True. If ϕ is True there, then it is trivially satisfiable; we may therefore assume $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

Although ϕ' is no longer a CNF, it can still be encoded as an MLP f via Lemma 8, so that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Under the assumption that a subset-minimal local δ -sufficient reason can be computed in polynomial time, we compute one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$ with $\delta := 1$, where the empirical distribution $\mathcal{D}$ is defined over the single data point $\mathbf{0}_n$.

We now claim that the subset-minimal δ -sufficient reason produced for $\langle f, \mathbf{x} \rangle$ spans the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Suppose first that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, fixing the complementary set to any value preserves the prediction; as the complementary set is $\emptyset$ here, this holds trivially. Subset-minimality of S means every $S' \subsetneq S$ satisfies $v(S') < \delta = 1$. Because the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by the single point $\mathbf{0}_n$, it takes only the values 1 or 0, so in fact $v(S') = 0$. Thus, apart from $S = \{1, \dots, n\}$ itself, no subset $S' \subsetneq S$ has the property that fixing the features in S' to 1 and the rest to 0 yields a classification of 1. Since the ϕ_2 component of ϕ' is True only when all features equal 1, this forces ϕ to be False on all of these inputs. Combined with the fact, verified at the outset, that ϕ is also False on $\mathbf{1}_n$, we conclude that $\phi \notin \text{CNF-SAT}$.

Conversely, suppose S is strictly contained in $\{1, \dots, n\}$. Since S is sufficient under our distribution $\mathcal{D}$ with $\delta = 1$, the same reasoning gives $v(S) = 1$, meaning that fixing the features in S to 1 and the rest to 0 makes f evaluate to 1. As this assignment is not the all-ones vector, the ϕ_2 component of ϕ' must be False, forcing the ϕ component to be True and hence $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

Having established the claim for neural networks, we extend it to tree ensembles with only a minor and straightforward adaptation, relying on the following lemma noted in several prior works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We reuse the very same construction as in the neural network proof, where ϕ' was encoded into an equivalent network. By construction ϕ' is no longer a valid CNF (though encoding it as an MLP posed no difficulty, since any Boolean circuit can be realized as an MLP). Because ϕ' is a conjunction of just two terms, however, it can be rewritten as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF and can be encoded as an equivalent random forest classifier, so the reduction described for MLPs carries over verbatim, completing the proof for both model families.

□

Lemma 9. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We give a proof analogous to that of Lemma 6, again reducing from the classical NP-hard CNF-SAT problem defined in Lemma 6. Specifically, given a Boolean formula ϕ , we show that computing a subset-minimal contrastive reason — for a neural network or a tree ensemble alike — decides the satisfiability of ϕ .

First, we check whether setting all variables of ϕ to 1 evaluates ϕ to True; if so, ϕ is satisfiable and we are done. Otherwise, we encode the CNF formula as a neural network f via Lemma 8. As in Lemma 6, we take $\mathcal{D}$ to be the empirical distribution supported on the single data point $\mathbf{0}_n$, set $\delta = 1$, and compute a subset-minimal δ -contrastive reason for $\langle f, \mathbf{x} \rangle$.

We now show that a subset-minimal δ contrastive reason exists for $\langle f, \mathbf{x} \rangle$ if and only if ϕ is satisfiable. The argument parallels that of Lemma 6. Because $\mathcal{D}$ is supported on a single datapoint, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is either 0 or 1, and since we seek a $\delta = 1$ contrastive reason, its existence is equivalent to the existence of an assignment that flips the classification of f — equivalently, one that changes ϕ from False to True. If such an assignment exists, ϕ is satisfiable. If no subset-minimal contrastive reason exists, then no subset of features fixed to zero (with the complementary set fixed to ones) yields a True output, which means no assignment evaluates f (and hence ϕ) to True, so ϕ is unsatisfiable.

To carry the proof over from neural networks to tree ensembles, we follow exactly the procedure of Lemma 6, encoding the CNF formula as an equivalent random forest classifier. The argument therefore holds for tree ensembles as well, concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We split the proof into two lemmas, one establishing the approximation guarantee and the other the absence of any bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We in fact prove a stronger, more general claim: the result holds for *any* model, subject only to the trivial condition that its inference runs in polynomial time and one further mild condition detailed below — both of which are satisfied by our neural network and tree ensemble formalizations.

We first note that, under empirical distributions, the global contrastive probability value function:

$$v_{con}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as already established in Proposition 2. Since Algorithm 2 issues only a linear number of such polynomial-time queries, its total runtime is polynomial.

For the approximation, the classical work of Wolsey et al. (Wolsey, 1982) gives a harmonic-series-based guarantee for integer-valued monotone non-decreasing submodular functions v , and more generally an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that v_{con}^g is monotone non-decreasing, and under feature independence Proposition 4 shows it is submodular. Together with the polynomial runtime of Algorithm 2, this immediately yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

It remains to show that $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is finite and bounded by $\mathcal{O}(\ln(|D|))$, hence effectively constant for a fixed dataset $|D|$. We first confirm the expression is well-defined and finite, then establish the bound. To this end we introduce a preprocessing step that removes “redundant” features — those that could in principle drive the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to zero. We begin by defining redundancy formally:

Definition 1. Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.

Here $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ denotes fixing all features in $[n] \setminus \{i\}$ to their values in $\mathbf{x}$ and feature i to its value in $\mathbf{z}$. This is equivalent to requiring:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes fixing the features in S to their values in $\mathbf{x}$ and those in $\bar{S}$ to their values in $\mathbf{z}$. In this sense, “removing” feature i from the input space and defining a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$ yields a function whose output on any input of size n' exactly matches that of f on the same features $[n] \setminus \{i\}$. This removal is polynomial-time for both of our model families: for neural networks it amounts to detaching the corresponding input neuron, and for tree ensembles to deleting every split on that feature across all decision trees. Owing to the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as noted earlier in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in D$ and testing whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds for some pair, feature i is not redundant; if it fails for every pair, i is redundant. Having identified all redundant features with respect to the empirical distribution $\mathcal{D}$, we remove them and build an equivalent model f' — a transformation that, as noted, is polynomial-time for both neural networks and decision trees.

To finish, note that $v_{\text{con}}^g(\{i\})$ equals 0 for an empirical distribution $\mathcal{D}$ precisely when feature i is redundant with respect to $\mathcal{D}$. Hence, once preprocessing has removed every redundant feature from f and produced f' , we are guaranteed $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now derive the sharper bounds promised above, showing that the approximation factor is $\mathcal{O}(\ln(|D|))$. As the empirical dataset D is fixed, this is a constant approximation. Tighter bounds may well be attainable, but our aim here is only to establish that the bound is *constant* — a fact that will stand in sharp contrast to the local setting, where no bounded approximation exists at all.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|D|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|D|^2}$. This in turn gives:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|D|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|D|^2) = \mathcal{O}(\ln(|D|)) \quad (71)$$

which concludes the proof.

□

Lemma 11. *Unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Suppose, for contradiction, that a bounded approximation algorithm exists for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, with f a neural network or a tree ensemble — even when $|D| = 1$. Lemma 9 shows, however, that even with a single baseline $\mathbf{z} = 0_n$ (i.e., a one-instance dataset), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. A polynomial-time bounded approximation for a cardinally minimal contrastive reason would implicitly decide this existence question, contradicting the lemma.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We split the proof into two lemmas: one establishing the approximation guarantee in the global case, the other the intractability of the local case.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof mirrors that of Lemma 10, where we showed that for both neural networks and tree ensembles Algorithm 2 attains an approximation factor of $\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right)$, which after preprocessing is finite and bounded by $\mathcal{O}(\ln(|D|))$.

As in Lemma 10, we again prove the stronger, more general claim that the result holds for *any* model, under the trivial condition of polynomial-time inference together with the redundant-feature removal condition described in Lemma 10 — both of which, as discussed there, are met by our neural network and tree ensemble formalizations.

Here too, since we work with empirical distributions, the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, whose approximation rested on the result of Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the supermodular nature of the function here calls for a different guarantee. Specifically, Shi et al. (Shi et al., 2021) provide an approximation factor of $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. We established in Proposition 2 that v_{suff}^g is monotone non-decreasing, and under feature independence Proposition 3 shows it is supermodular. Combined with the polynomial runtime of Algorithm 2, this yields an approximation guarantee of $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus i)}{v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset)}$.

To show this expression is bounded and constant, we apply the same preprocessing as in Lemma 10, removing all redundant features from f — a step that, as explained there, is polynomial-time for both neural networks and tree ensembles. As before, a feature i is strictly redundant iff $v(\{i\}) - v(\emptyset) = 0$. This yields a new function f' behaving identically to f on the remaining features and guaranteeing that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$.

To show that this expression is both bounded and constant, we carry out — exactly as in Lemma 10, and again in polynomial time for both neural networks and tree ensembles — the same preprocessing phase that strips f of all redundant features. Here too, a feature i is strictly redundant iff $v(\{i\}) - v(\emptyset) = 0$. The preprocessing thus produces a new function f' that both reproduces the behavior of f on all remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, by the same reasoning as in Lemma 10 — where the probability term is lower bounded by $\frac{1}{|D|}$ and its expectation by $\frac{1}{|D|^2}$ — we obtain $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. As in Lemma 10, the first term is then bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we argue as follows:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$, whence $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore $1 - |D|^2 \leq k^f \leq 0$. This gives $\frac{1}{1 - k^f} \leq |D|^2$, so the overall approximation bound derived from Shi et al. is constant. This bound may be much smaller in practice, but here we only need it to be constant — again in stark contrast to the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof parallels Lemma 11. Suppose, for contradiction, that a polynomial-time algorithm provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, with f a neural network or a tree ensemble — even when $|D| = 1$. Lemma 6 shows that even in the extreme single-baseline case $\mathbf{z} = 0_n$, deciding whether a sufficient reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Such an approximation algorithm would decide this existence question, contradicting the hardness result.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds via a polynomial-time, approximation-preserving reduction from a variant of Set Cover that, like ordinary Set Cover, is provably hard to approximate within the factor stated in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ over a dataset $\mathbf{D}$ and threshold δ' . We first recall the

definition of a global contrastive reason and then specialize it to the case where the explanation is evaluated over an empirical dataset $\mathbf{D}$:

$$\mathbb{E}_{\mathbf{x} \sim \mathbf{D}}[\Pr_{\mathbf{z} \sim \mathbf{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathbf{D}}[\Pr_{\mathbf{z} \sim \mathbf{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathbf{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) & \end{aligned} \quad (74)$$

where $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f , first specifying the functional form it should implement. For each set S_j we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both in $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, agreeing on every coordinate outside S_j — that is, sharing identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by drawing them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$, so that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = \mathbf{y}^j \quad (76)$$

We aim to construct f so as to satisfy:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}^j) \quad (77)$$

Assume now that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — i.e., no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption we obtain:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathbf{D}}[\Pr_{\mathbf{z} \sim \mathbf{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) &= \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \end{aligned} \quad (78)$$

Here the factor of 2 in both numerator and denominator arises because only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for every S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total this produces $2m$ vectors, which form the dataset $\mathbf{D}$ from which we build f .

We then construct a neural network f realizing this function, using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network representing any finite set of input–output pairs can be built in polynomial time. We construct f accordingly and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

We now argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Since each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, these pairs never overlap, so the coverage of any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. Before giving the full proof, we recall the definition of orthogonal DNFs (Crama & Hammer, 2011), the broader model class to which we extend our decision-tree guarantee.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This means that no single variable assignment can satisfy more than one term simultaneously.

The proof mirrors the argument of Theorem 1, resting on the fact that v_{suff}^g is polynomial-time computable for any set S in this broader class as well.

For a given set S , let $\mathcal{C}^S T_i$ denote the constraints imposed by the term T_i on the variables in S . We enumerate all pairs of terms (T_i, T_j) and discard any pair whose constraints on S are *inconsistent* (i.e., $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$), since such a pair contributes zero to v_{suff}^g .

For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$. The contribution of this pair to v_{suff}^g is then

$$\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. As there are at most m^2 such term pairs, the value v_{suff}^g is computable in polynomial time under an independent distribution $\mathcal{D}$, extending the tractability guarantee from decision trees to this more general class. $\square$

O DISCLOSURE: USAGE OF LLMs

An LLM was used exclusively as a writing assistant to refine grammar and typos and improve clarity. It did not contribute to the generation of research ideas, study design, data analysis, or interpretation of results, all of which were carried out solely by the authors.